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Multi-Segmented Drainage Volume Analysis and Forecasting with Parent-Child Interactions

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Abstract

Dynamic drainage volume (DDV) in unconventional reservoirs can be estimated from diffusive time-of-flight analysis using well rates and pressures. However, in multistage fractured horizontal wells, production is often influenced by inter-well communication, particularly frac-hits from child well completions near parent wells. These events can significantly alter well productivity, making accurate detection and quantification essential for performance monitoring, forecasting, and optimization. This paper presents a hybrid, physics-informed, data-driven approach—multi-segmented dynamic drainage volume (ms-DDV)—to continuously estimate DDV in the presence of well interference and forecast its impact on well performance.

The ms-DDV approach extends single-well DDV calculations to handle performance discontinuities by segmenting the drainage volume evolution and applying a hybrid SINDy (Sparse Identification of Nonlinear Dynamics) framework. This method automatically detects events such as frac-hits, refracs, or stimulations, quantifies their effects on drainage volume, reservoir pressure, and productivity index (PI), and uses the latest segment for accurate condition-based forecasting. Field case studies demonstrate the method's capability to assess both positive and negative impacts of interference events, enabling reliable PI-based rate forecasts and dynamic inflow performance relationship (IPR) computations. By combining the rigor of physics-based modeling with the adaptability of data-driven techniques, ms-DDV offers a robust, field-proven tool for interpreting frac-hit impacts and integrating these insights into production optimization workflows across U.S. onshore basins.

Introduction

The shale revolution and the widespread adoption of multi-stage hydraulic fracturing have unlocked large volumes of hydrocarbons from ultra-low permeability formations by creating extended stimulated reservoir volumes and enabling production from previously inaccessible rock (Horne and Temeng, 1995; Wan and Aziz, 1999; Arthur et al., 2009). At the same time, the closely spaced development patterns common in many onshore U.S. plays have introduced a practical challenge: parent-child interference. When a newly completed (child) well is placed near an existing (parent) well, induced fractures can intersect or communicate with the parent's stimulation and drainage region, producing so-called "frac-hits." These

interactions can either temporarily boost parent-well productivity (e.g., from invading fracturing fluid and re-pressurization) or cause permanent reductions in deliverability because of shared drainage, accelerated depletion, or mechanical damage (Ibrahim et al., 2021; Joe and Sharma, 2022). Detecting, quantifying, and forecasting the consequences of such interference remains a pressing problem for field development planning, real-time operations, and reservoir management.

Modeling and forecasting performance in unconventional reservoirs is inherently difficult. The prolonged transient-flow behavior and evolving drainage volume in ultra-low permeability reservoirs undermine the simplifying assumptions used in traditional analytical methods and decline-curve extrapolation (Wattenbarger et al., 1998; Bello and Wattenbarger, 2010). High-fidelity physics-based simulation can in principle capture the relevant mechanisms, but it requires detailed knowledge of spatial rock and fracture properties that are typically unavailable at field scale. Conversely, purely empirical approaches such as Arps decline-curve analysis are simple and data-efficient but lack the physics to accommodate surface changes, artificial lift alterations, or unmodeled phenomena like well interference (Arps, 1945). Midway approaches—Rate Transient Analysis (RTA) and PI-based forecasting—bring more physics into data-driven workflows but remain limited by interpretive manual steps, restrictive flow-regime assumptions, and sensitivity to incomplete inputs (Bello and Wattenbarger, 2008; Clarkson, 2021).

One particularly useful physics-aware quantity for unconventional wells is the dynamic drainage volume (DDV), which characterizes the time-evolving region of reservoir that contributes to production and can be estimated from diffusive time-of-flight using rate and pressure time series. Existing DDV formulations and single-well analyses, however, are not designed to handle abrupt discontinuities in well performance caused by frac-hits, refracs, mechanical events, or other operational perturbations. Current detection practices for interference—manual inspection of pressure spikes, water-cut anomalies, or stage-by-stage analysis of injection/leak-off data—are often ad hoc, labor-intensive, and inconsistent across fields (Ibrahim et al., 2021; Rahman et al., 2021). Importantly, pressure signatures do not always correlate directly with production impact, so a detection metric that links changes in drainage volume to inflow performance and forecasting is desirable (Daneshy, 2014, 2018; Sahai et al., 2015; Wang et al., 2021).

In this work we extend the DDV concept to explicitly account for discontinuities in well behavior by introducing a multi-segmented dynamic drainage volume (ms-DDV) formulation and a hybrid modeling framework that fuses data-driven identification with physics-based PI (productivity index) principles. The central idea is to partition a well's production history into segments defined by changes in drainage response, automatically identify governing dynamics for each segment using sparse system identification (SINDy), and use the most recent segment to produce forecasts conditioned on the current reservoir state. This approach provides three practical advantages: (1) continuous, automated detection of interference and other anomalies via changes in the estimated drainage volume; (2) interpretable, parsimonious dynamic models for each segment that capture the dominant flow relationships; and (3) PI-consistent forecasts and dynamic IPR estimates that reflect the altered drainage and inflow conditions following an event.

The proposed ms-DDV workflow builds on prior PI-based forecasting and hybrid RTA ideas (Molinari and Sankaran, 2021; Sinha et al., 2023; Zalavadia et al. 2024) but introduces two key innovations. First, the segmentation of drainage volume explicitly represents discontinuities due to frac-hits, refracs, stimulation, or operational changes and provides a natural metric for quantifying the intensity and sign (positive or negative) of interference. Second, the use of Sparse Identification of Nonlinear Dynamics (SINDy) enables discovery of compact mathematical descriptions of the segment-level flow regime dynamics for dynamic drainage volume estimates directly from routinely available inputs (rates, wellhead pressure, minimal PVT and wellbore data). The result is a hybrid, physically grounded yet data-efficient framework that is both field-deployable and scalable across many wells.

This paper makes the following key contributions and enhancements:

1. Hybrid SINDy modeling: an integrated modeling pipeline that uses sparse identification to learn interpretable segment dynamics based on dominant flow regimes and couples these with PI-based forecasting to produce rate forecasts and dynamic IPRs consistent with the current drainage volume state.
2. Multi-segmented DDV formulation (ms-DDV): a robust extension of single-well DDV that handles discontinuities in drainage behavior and provides continuous estimates of segment wise drainage volumes and their evolution.
3. Automated detection and quantification of interference: an algorithmic workflow to identify breakpoints, classify events (e.g., frac-hit, refrac), and quantify impacts on cumulative production and productivity indices.
4. Field demonstrations: multiple case studies from U.S. onshore basins showing the method's ability to detect both positive and negative interference effects on drainage volume access and PI of the well, to improve forecast accuracy across event boundaries, and to provide transient PI estimates useful for downstream optimization.

The remainder of the paper is organized as follows. Section 2 reviews background on DDV, PI-based forecasting, and sparse system identification. Section 3 presents the ms-DDV formulation and the hybrid SINDy–PI workflow, including breakpoint detection, segmentation logic, and forecasting procedures. Section 4 demonstrates the approach on several field case studies and compares ms-DDV forecasts to conventional methods. Section 5 discusses practical considerations, limitations, and extensions (including multiphase effects and field-scale aggregation). Finally, Section 6 summarizes conclusions and outlines future work for integrating ms-DDV into operational production-forecasting pipelines.

Methodology

The proposed framework introduces a hybrid reservoir modeling approach for Transient Well Performance (TWP), designed to enhance production diagnostics and forecasting. The formulation integrates three key components: diffusive time-of-flight calculations, sequential pseudo–steady-state (SPSS) material balance principles, and transient productivity index (PI) concepts (Molinari and Sankaran 2021). This approach has been successfully deployed across thousands of wells in multiple shale plays and tight reservoirs in North American.

The foundation of TWP lies in the concept that drainage volume evolves over time without a fixed geometry. The diffusive time-of-flight method is employed to estimate this expanding drainage volume under a no-flow boundary condition, with cumulative production data used to account for reservoir withdrawals. Within the SPSS framework, high-resolution material balance calculations (e.g., on a daily basis) provide dynamic estimates of reservoir pressure depletion and average pressure within the contacted volume.

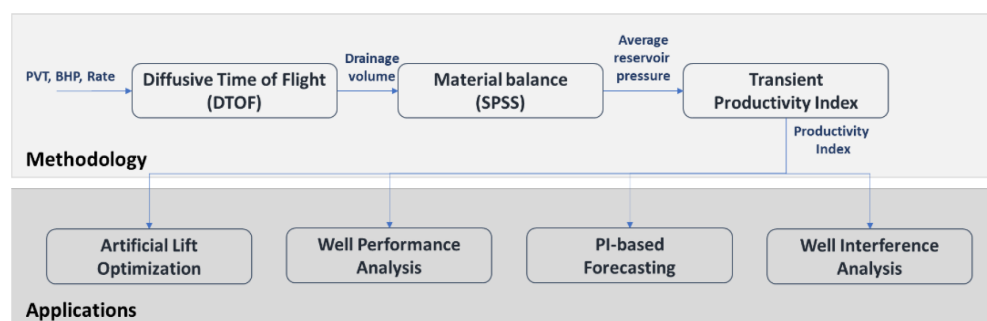


Figure 1—Transient Well Performance (TWP) methodology and selected applications

Applications of TWP

TWP utilizes routinely available production data, in conjunction with pressure–volume–temperature (PVT) properties and flowing bottomhole pressure measurements (Molinari and Sankaran 2021). The framework is capable of handling variable production rates, reservoir pressure depletion, nonlinear pressure-dependent PVT behavior, and multiphase flow under different artificial lift systems such as natural flow, gas lift, electrical submersible pumps (ESP), and sucker rod pumps. Insights derived from TWP support a range of production engineering applications, including inflow/outflow opportunity assessment, anomaly detection, PI-based forecasting, artificial lift optimization, and scenario modeling.

Drainage Volume Estimation

Dynamic Drainage Volume (DDV) is estimated using pressure and rate data as follows:

$$V_d \approx \frac{1}{c_{ti} \frac{d}{dt_e}(RNP)} \quad \text{Equation 1}$$

This formulation represents the effective contacted pore volume at any given time step, tracking the extent of pressure propagation within the reservoir. Like the concept of the radius of investigation in homogeneous formations, it delineates the diffusive time-of-flight contours within an irregularly shaped drainage area, encompassing both the stimulated and unstimulated reservoir matrix. The Rate Normalized Pressure (RNP)-based calculation offers a representative model of production behavior under a standardized reference rate. An enhancement to the previous approach laid in this paper is around calculation of RNP derivative required in this formulation. In order to calculate the derivative of RNP, our approach is centered on estimating RNP (rate normalized pressure) as a function of a set of superposition time-dependent basis functions for primary phase rate. Material balance time (MBT) is the superposition time function used here that normalizes variable-rate production data to resemble a constant-rate solution as typically a routine followed in RTA. The SINDy formulation is represented as follows:

$$RNP = \Theta(MBT)E \quad \text{Equation 2}$$

$$RNP = \sum_i a_i t_e^{m_i} \quad \text{Equation 3}$$

$$RNP' = \sum_i a_i (m_i - 1) t_e^{m_i - 1} \quad \text{Equation 4}$$

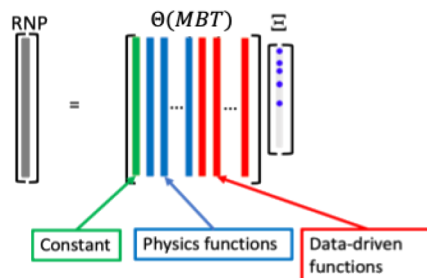


Figure 2—SINDy methodology

Here, $\Theta(MBT)$ represent a library of MBT basis functions and E represents the coefficients for each basis function. In hydraulically fractured wells, reservoir and fracture properties are typically uncertain, and conventional rate-transient analysis (RTA) often relies on straight-line diagnostic techniques derived from characteristic flow equations. While useful, such approximations depend on simplifying assumptions about fracture mechanics and reservoir geometry and may involve manual interpretation, which introduces subjectivity. An alternative is to adopt a data-driven framework, such as the Sparse Identification of Nonlinear Dynamics (SINDy) method, to derive governing equations of fluid flow directly from commonly

available measurements (e.g., rates and pressures). By leveraging the information embedded in production data, this approach aims to identify the dominant features that control well and reservoir performance. The methodology incorporates a library of time-dependent basis functions informed by fluid flow physics in fractured reservoirs, enabling SINDy to recover a parsimonious representation that highlights the key factors influencing observed rate–pressure behavior. With this analytical estimate of RNP in terms of material balance time, we can employ the obtained equation to calculate the derivative required for dynamic drainage volume (DDV) estimate. The RNP vs MBT plot is often noisy when dealing with real world dataset and hence can lead to inaccuracies when fitting the proposed SINDy model. [Chu et al. \(2019\)](#) suggested the use of the integral RNP with respect to time (IRNP) and its derivative for the study of flow-regime signatures in the Wolfcamp shale to handle noisy data. The same approach is applied using IRNP in [Equation 5](#) for training the SINDy model. Thus, [Equation 5](#) can be easily modified to IRNP form and the coefficients obtained by SINDy can be converted to coefficients corresponding to RNP and used in the DDV equation.

$$IRNP = \frac{1}{t_e} \int_0^{t_e} \frac{p_i - p_{wf}(\tau_e)}{q(\tau_e)} d\tau_e = 1 \left| t_e \int_0^{t_e} RNP(\tau_e) d\tau_e \right. \quad \text{Equation 5}$$

Table 1—Flow regimes features

Flow Regimes	Features
Bilinear flow	$t^{0.25}$
Linear flow to individual fractures	$t^{0.5}$
SRV / PSS flow	t
Compound linear flow	$t^{0.5}$
Sub-linear flow $\delta \in (0, 0.5)$, Sub-radial flow $\delta \in (0.5, 1)$	$t^{1-\delta}$

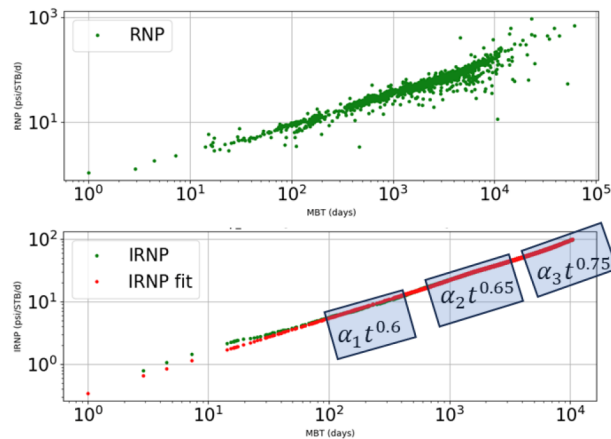


Figure 3—RNP vs. MBT and IRNP vs. MBT (actual and fit) showing how SINDy would detect the dominant flow regimes with time

The proposed approach contains two main steps, i.e., breakpoint identification and segment verification show in [Figure 4](#) as below:

1. Breakpoint identification: During normal production, the pressure normalized rate of the well (PNR) exhibits a continuous decline in transient flow. However, when a production interference event occurs, often caused by offset well fracturing, significant changes in the PNR decline trend can be observed with a sudden PNR increase typically at event onset. During a frac-hit event, a sharp influx of water in the well is observed leading to increased watercut. In the first step, any breakpoints in the PNR decline,

a change in the PNR trend and abrupt watercut increase are automatically detected and defined as segments.

2. Segment verification: Subsequently, it is crucial to verify whether the segmented PNR decline is a result of a frac-hit or refrac event from the production of nearby offset wells and completion data.

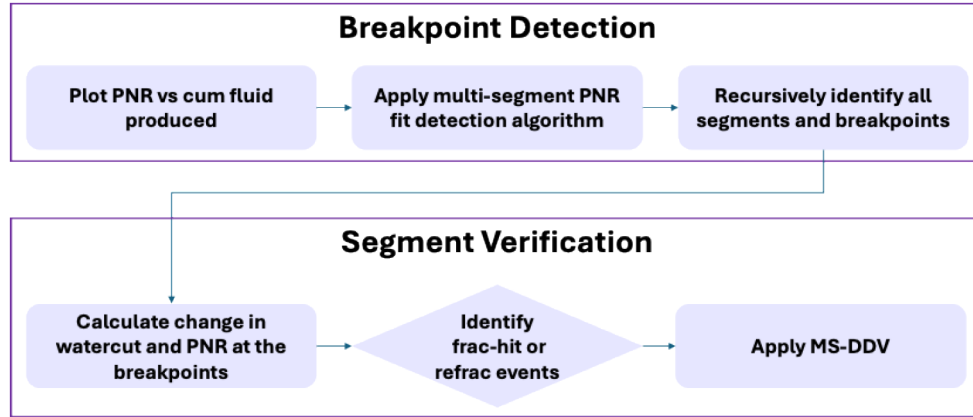


Figure 4—Segment detection workflow for ms-DDV

Once the segments are detected, the SINDy based approach highlighted above is applied to individual segments. The change in the DDV at the onset of each event/segment is calculated using equation below:

$$\Delta DDV = \frac{1}{c_{ti}} \left(\frac{1}{\frac{dRNP}{dMBT}_{post-event}} - \frac{1}{\frac{dRNP}{dMBT}_{pre-event}} \right) \quad \text{Equation 6}$$

where $\frac{dRNP}{dMBT}$ is computed at same MBT on event onset, by extrapolating the MBT trend on one of the segments. A similar approach to predict change in the contacted fluid in place is described in Clarkson et al. 2024. Figure 5 shows the detected breakpoint in PNR using the algorithm described above that checks for a new PNR trend and an abrupt increase in watercut. The change in DDV at the breakpoint is shown in Figure 5 followed by which the DDV profile has a slower rate of increase.

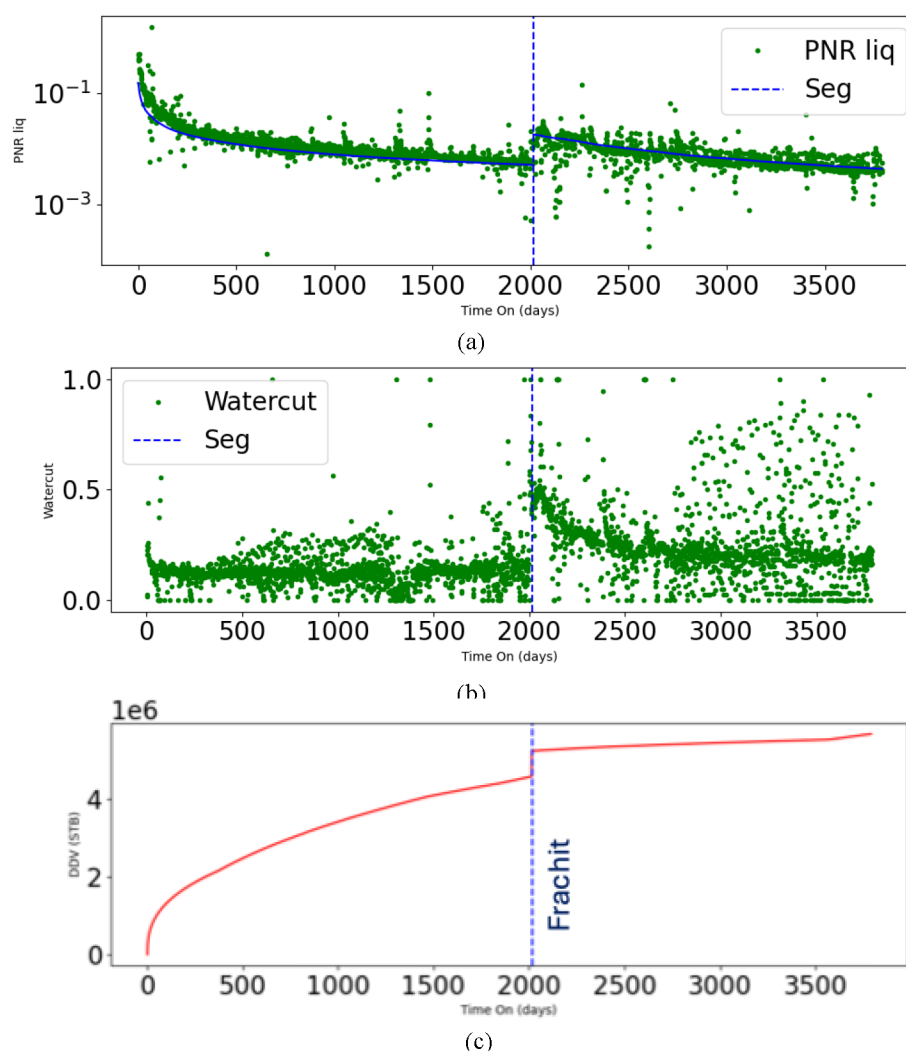


Figure 5—Example well that suffered a frac-hit around 2000 days (blue line) into production with (a) PNR liquid showing a different trend post the event and a sudden jump at event onset and (b) Watercut showing a sudden increase at event onset (c) DDV profile as computed using the formulation stated using SINDy for RNP derivative calculation.

Material Balance with an Expanding Control Volume

Traditional material balance methods are strictly applicable in steady-state or pseudo-steady-state conditions, where reservoir boundaries have been fully contacted. However, unconventional reservoirs seldom reach boundary-dominated flow due to their low permeability, where even after fracture connectivity is established, additional matrix volume continues to contribute to flow over time.

To address this, an expanding control volume approach is implemented. This method conceptualizes reservoir drainage as an evolving boundary, analogous to an expanding balloon, where pressure contours define the outer boundary's progression. The SPSS framework approximates transient flow as a sequence of pseudo-steady-state snapshots, with drainage volume computed at each time step as shown in Figure 6. Given that the system is defined by a no-flow (DDV concept), outer boundary at each instant, energy loss due to production can be directly estimated, allowing for accurate reservoir depletion calculations. The use of daily production increments enhances the precision of material balance assessments.

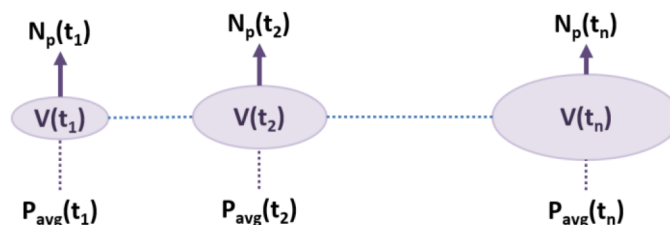


Figure 6—Material Balance applied to succession of pseudo-steady-state conditions with expanding drainage volume

For liquid systems, the general material balance equation accounts for the following four states: liquid expansion, liberated gas expansion, change in pore volume due to connate water/residual oil expansion and pore volume reduction, and underground withdrawal. By applying first principles, this equation is adapted to unconventional oil reservoirs lacking a free gas cap or active aquifer support. The equations for the proposed states are:

$$\text{Oil expansion term:} \quad N(B_o - B_{oi}) \quad \text{Equation 7}$$

$$\text{Liberated gas expansion term:} \quad N(R_{si} - R_s)B_g \quad \text{Equation 8}$$

$$\text{Expansion of water and PV reduction:} \quad dV_w - dPV = (NB_{oi})\left(\frac{S_w c_w + c_F}{1 - S_w}\right)\Delta p \quad \text{Equation 9}$$

$$\text{Underground withdrawal:} \quad N_p[B_o + (R_p - R_s)B_g] + W_p B_w \quad \text{Equation 10}$$

Then, using the dynamic drainage volume the oil in place (N), can be estimated as Equation 11, and the final equation is Equation 12.

$$V_d = \frac{N}{1 - S_w} \quad \text{Equation 11}$$

$$R_p = R_s + \frac{V_d(1 - S_w)}{B_g N_p} \left[(B_o - B_{oi}) + (R_{si} - R_s)B_g + B_{oi} \left(\frac{S_w c_w + c_F}{1 - S_w} \right) \Delta p \right] - \frac{B_o}{B_g} - \frac{W_p B_w}{N_p B_g} \quad \text{Equation 12}$$

Using the above set of equations, an optimization procedure is followed to obtain the average reservoir pressure profile that best matches the predicted cumulative GOR (R_p using the material balance equation stated above) with the actual historical cumulative GOR trend. Similar set of equations can be applied for gas wells.

Transient Productivity Index

The productivity index (PI) is a widely recognized diagnostic metric that characterizes reservoir inflow potential by relating production volumes to flowing pressure conditions. Unlike rate–time–based decline curve analysis (DCA) or type-curve methods, PI inherently accounts for depletion effects and pressure–volume–temperature (PVT) behavior, thereby reducing biases associated with variable operating conditions.

In unconventional reservoirs, PI exhibits a transient nature and must be updated continuously to capture dynamic reservoir behavior. By linking production rates with pressure drawdown, PI provides a quantitative framework for evaluating well performance and supports optimization of operational strategies such as artificial lift selection, production adjustments, and broader reservoir management practices.

$$q_f(t) = PI(t) * (p_{avg}(t) - p_{wf}(t)) \quad \text{Equation 13}$$

As the average reservoir pressure and pressure drawdown behavior is calculated by material balance and drainage volume concept (shown in Figure 7 (a), (b), (c)), the dynamic PI be then calculated using daily production data as shown in Figure 7 (d)

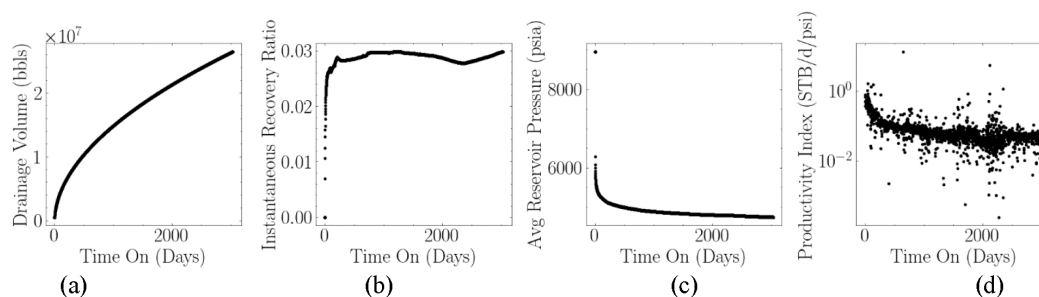


Figure 7—(a) Drainage volume, (b) instantaneous recovery factor, (c) average reservoir pressure and (d) transient productivity index for a well from a liquid-rich US shale play

Productivity-based Forecasting (PIBF)

Traditional decline curve analysis (DCA) does not explicitly incorporate pressure data or fluid phase behavior and is best suited for stable operating conditions, such as constant bottomhole pressure or fixed choke settings with minimal interruptions. In practice, however, DCA may be unreliable or even inapplicable when production is constrained by facilities or operational limits, which can obscure natural decline trends.

To overcome these limitations, the productivity index (PI) can be adopted as the primary forecasting variable. PI inherently accounts for pressure effects and phase behavior, reducing noise in the data and producing clearer decline trends that require fewer data points for calibration. Empirical studies across multiple shale and tight reservoir settings have validated the robustness of PI trends, even under constrained production conditions. Unlike rate-based methods that normalize pressure, PI incorporates both depletion and PVT effects, offering a more consistent diagnostic and forecasting basis (Molinari and Sankaran, 2021).

PI trends can be modeled using a variety of time-series techniques, including modified hyperbolic models, ARIMA, and recurrent neural networks (RNNs), applied to historical production and cumulative volumes. The framework assumes that wells remain on primary depletion without abrupt productivity changes such as refracturing or severe interference. Importantly, well interference events are often more discernible in PI trends than in raw rate or pressure data, as shifts in PI can clearly indicate external influences (Rahman et al., 2019). Figure 8 illustrates a shift in PI trends at approximately 400 and 2300 days, corresponding to the completion of a nearby child well and subsequent changes in production behavior. In such cases, multi-segment PI-based forecasting (MS-PIBF) can be employed, where distinct PI segments are fitted before and after interference events to capture changes in decline behavior (Sinha et al. 2023). This enables quantification of parent-child well interactions, which may accelerate depletion or, in some cases, temporarily enhance parent well productivity due to effects such as fracture communication, fluid pressurization, or inadvertent restimulation.

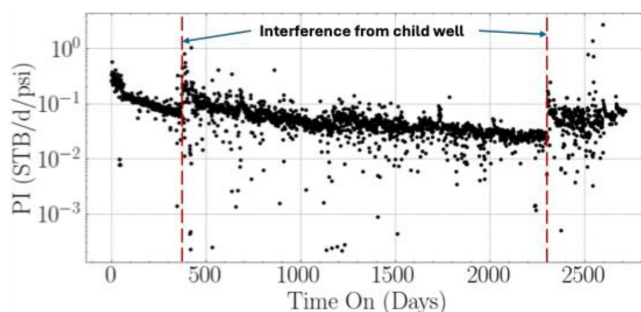


Figure 8—Example of well interference identified by PI after 400 days and 2300 days due to child well completion and production

The PI forecast can be transformed into liquid production by combining it with reservoir pressure projections (from material balance relationships) and bottomhole pressure forecasts (based on choke schedules, artificial lift, or other operating strategies). Once single-phase liquid forecasts are established, multiphase profiles—oil, gas, and water—are generated by modeling water cut and gas–oil ratio (GOR) (Zalavadia et al. 2022). GOR evolution is influenced by factors such as flowing BHP, saturation pressure, PVT variations, pressure-dependent permeability, fracture geometry, and relative permeability trends (Pradhan, 2020).

By integrating these considerations, the PI-based forecasting framework provides a robust and flexible methodology for predicting well performance under dynamic operating conditions while explicitly accounting for reservoir behavior, operational changes, and well interactions.

Result and Discussion: Case Study

To evaluate the effectiveness of the proposed Multi-Segmented Drainage Volume (ms-DDV) Analysis and Forecasting framework in handling parent–child well interactions, five representative field case studies were analyzed. These examples illustrate the method's ability to detect and quantify both beneficial and detrimental impacts of frac-hit events on dynamic drainage volume (DDV), reservoir pressure, and well productivity. The cases span a range of responses, demonstrating that frac-hit effects are not universally negative—depending on reservoir connectivity and stress redistribution, they can either enhance or impair well performance. The ms-DDV approach successfully characterizes these impacts in real time and provides actionable inputs for operational decision-making and forecasting.

Case Study 1 – Average Reservoir Pressure and Production Hindcast Validation

The Transient Well Performance (TWP) methodology was first validated using wells equipped with downhole pressure gauges. Calculated average reservoir pressures from the TWP analysis were compared against measured long shut-in gauge pressures which are rare to have in unconventional, however, in a few instances these were available. As shown in Table 2, the calculated values closely matched the measured data, with deviations typically within $\pm 7\%$, confirming that the TWP-derived pressures provide a reliable representation of in-situ reservoir conditions. Two representative examples are presented in Figure 9, where the blue box indicates the shut-in period. In both cases, the downhole gauge pressure (grey line) gradually increased during shut-in and converged toward the calculated reservoir pressure (blue line). Given the slow propagation of pressure transients in unconventional reservoirs, it is expected that extending the shut-in duration would result in even closer agreement between the measured and calculated pressures. This behavior further validates the effectiveness of the TWP approach for reservoir pressure estimation in this study.

Table 2—Reservoir pressure validation by gauge pressure measurement in shut-in period

	Well 1	Well 2	Well 3	Well 4	Well 5	Well 6
Calc. Reservoir Pr. (psi)	4426	4081	3805	5310	4425	3881
Meas. Gauge Pr. (psi)	4264	3966	3691	4755	4958	3760



Figure 9—Example of reservoir pressure validation using shut-in gauge pressure measurement

The TWP model was further validated by hindcasting cumulative oil and liquid production and comparing it with historical production data. This exercise used the exact daily BHP (Pwf) during the hindcast period and fitted PI trend using Arps hyperbolic decline. Model performance was assessed using Mean Absolute Percentage Error (MAPE). As summarized in Table 3, the hindcast results exhibited strong agreement with the actual production history, with average MAPE values of 13.3% for oil and 11.3% for liquids across the 14 wells analyzed. These results confirm that the mutli-segmented DDV approach can provide accurate estimates of average reservoir pressure and reliably reproduce historical production trends. This verification step establishes a solid baseline for the subsequent multi-segment TWP analysis presented in Section 3.2.

Table 3—TWP validation MAPE results

Well	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Avg
Cum Oil MAPE (%)	16.1	11.1	5.4	12.3	18.9	20.9	11.4	14.2	2.2	22.0	34.5	1.3	9.8	6.7	13.3
Cum Liq MAPE (%)	13.4	5.8	15.8	1.7	22.1	25.9	7.5	14.7	2.4	15.4	2.1	0.3	26.3	5.3	11.3

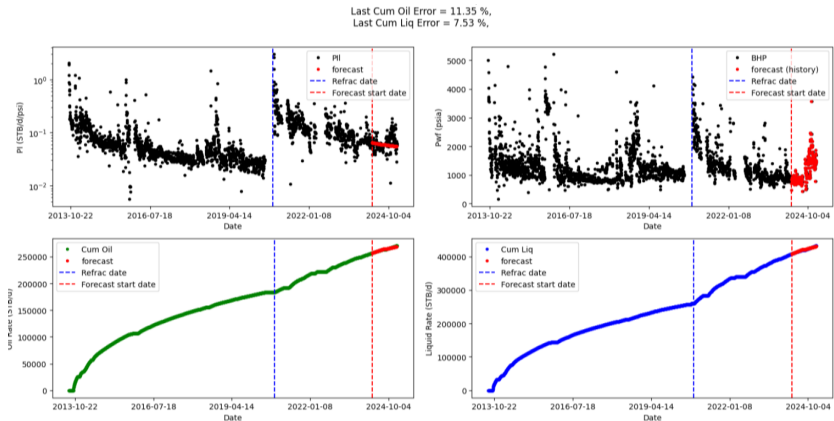


Figure 10—TWP validation by hindcasting cumulative oil and liquid production volume

Case Study 2 – Impact of Frac-hit Event on Dynamic Drainage Volume Estimate (Decrease in DDV)

The MS-DDV analysis captured a sharp decrease in dynamic drainage volume (Figure 11) consistent with a frac-hit originating from neighboring well. This event, occurring around day 1,950, was clearly identified by the segmentation (dashed blue line) and was accompanied by a sudden drop in the calculated average reservoir pressure (green curve in the top-right panel). The simultaneous decrease in both DDV and average pressure suggests a loss of connected drainage area, likely due to fracture reorientation or stress redistribution caused by the interference event where the child well takes up some of the parent well's drainage volume. Bottom-hole pressure (blue dots) also showed a rapid decline immediately following the event, indicating reduced reservoir support and altered flow behavior.

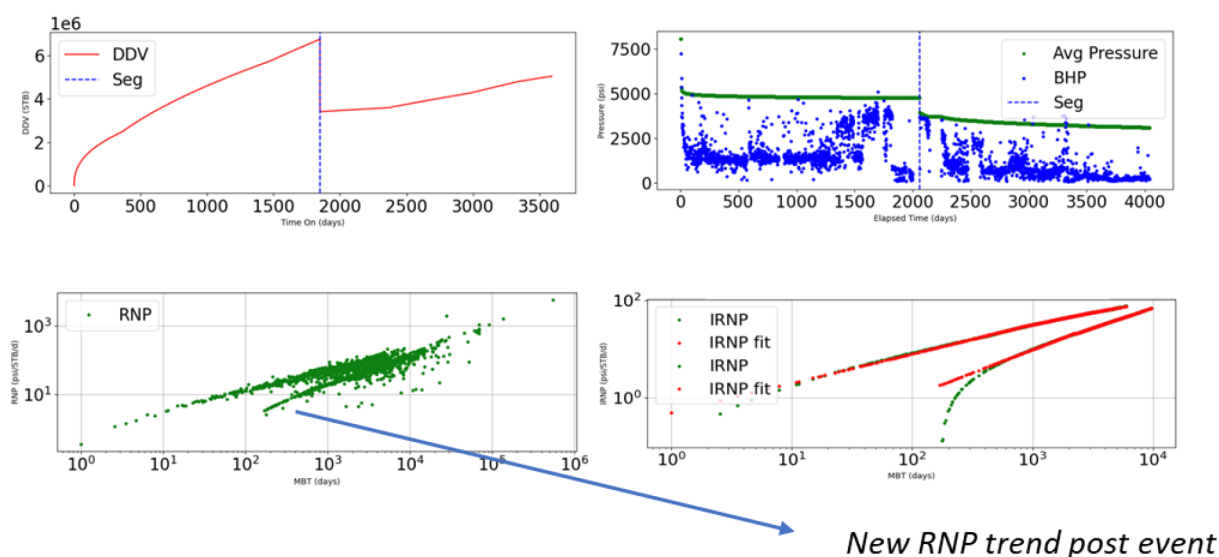


Figure 11—Example: frac-hit leading to reduction in DDV

The rate-normalized pressure (RNP) diagnostic (bottom-left) exhibited a break in trend post-event, highlighting a change in the reservoir's flow capacity. This was further confirmed by the integral RNP (IRNP) analysis (bottom-right), where the fitted curves for the post-event segment showed a significantly steeper slope compared to the pre-event segment, implying a reduction in productivity index (PI) and accelerated pressure depletion due to the boundary dominated flow observed by the well. Together, these multi-diagnostic indicators confirm that the frac-hit had a detrimental effect on well deliverability, shortening the effective life of the current drainage volume. Importantly, the MS-DDV framework not only identified the timing and magnitude of the event but also provided a quantitative basis for forecasting the post-interference performance trajectory.

Case Study 3 – Impact of Frac-hit Event on Dynamic Drainage Volume Estimate (Increase in DDV)

Figure 12 illustrates a well that experienced a frac-hit from nearby stimulation activity, which in this case had a **positive** impact on reservoir connectivity. The MS-DDV analysis shows a clear upward shift in dynamic drainage volume (top-left) following the event around day 2,000, as identified by the segmentation marker (dashed blue line). This expansion in DDV is likely due to newly connected reservoir volume, from fracture network extension or reactivation induced by the offset well's stimulation. Also, the rate of increase in DDV post frac-hit remains similar as pre-event.

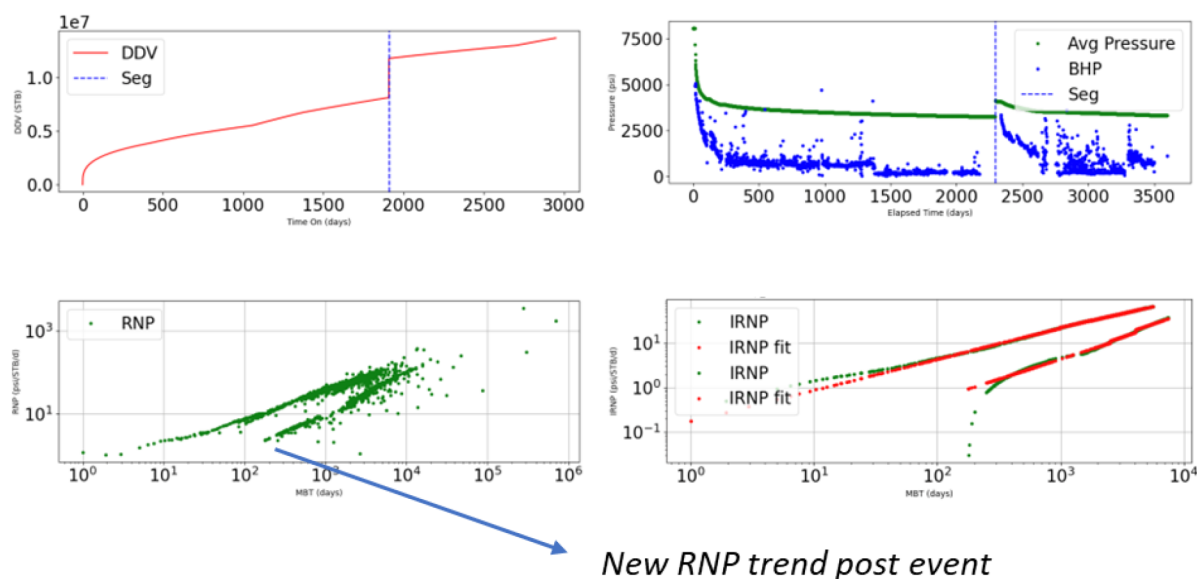


Figure 12—Example: frac-hit leading to increase in DDV

The average reservoir pressure (green curve, top-right) as a result of higher drainage volume also exhibited a sharp increase, indicating that the event energized the pore volume introduced by higher-pressure fluid into the connected system. The integral RNP (IRNP, bottom-right) shows close to linear flow regime trend even after frac-hit events helping with better productivity as it accesses more drainage volume. This case demonstrates that frac-hits, when favorably oriented and connected, can have a measurable and lasting positive effect on well performance, and that the MS-DDV framework provides a robust means of quantifying such benefits for forecasting and field optimization.

Case Study 4 – Frac-hit Impact Quantification (Positive Impact)

In this example, both the Productivity Index Based Forecasting (PIBF) and the Multi-Segmented PIBF (MS-PIBF) methods were trained until 90 days before the last production data, followed by a blind validation on the last 90 days. The MS-PIBF results are shown Figure 13, showing very good agreement against actual daily trends confirming the accuracy of multi segmented forecasts. The MS-PIBF approach is shown to consistently outperform the single-segment PIBF in capturing the production dynamics across the interference event.

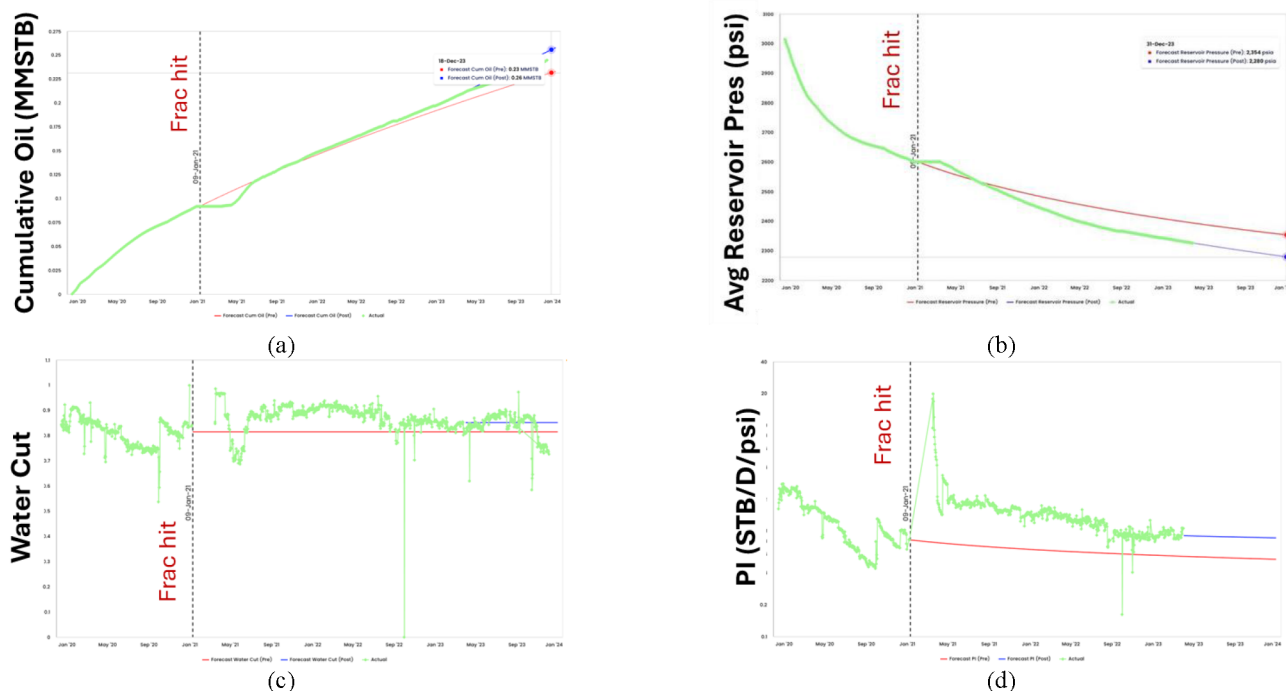


Figure 13—Single interference event (positive impact) – Multi segmented forecast of production rate, liquid productivity index, flowing bottom hole pressure and water cut

Following validation, forward forecasts for PI, bottomhole pressure (BHP), and associated fluid production rates were generated for up to one year beyond the last production date. Figure 13 illustrates the segmented forecasts, distinguishing between pre-frac-hit (Segment 0) and post-frac-hit (Segment 1) trends. Table 4 summarizes the forecasted cumulative oil production for each segment, showing an increase from 0.27 MMSTB pre-event to 0.42 MMSTB post-event, a net gain of 0.15 MMSTB.

Table 4—Single interference event (positive impact – Cumulative oil forward forecast for each segment (pre-frac hit and post-frac hit) using MS-PIBF

	Forecasted Cumulative Oil Volume (MMSTB)
Pre-Event	0.27
Post-Event	0.42
Difference	0.15

This positive impact is attributed to the frac-hit event creating a high-pressure zone within the parent well's drainage area. The injected frac fluid from the child well not only re-pressurized the local reservoir but also mobilized additional hydrocarbons, temporarily increasing both productivity and short-term recovery factor. This case demonstrates the MS-PIBF method's capability to quantify such beneficial interference events and to provide a more accurate forecast by incorporating event-driven performance shifts into the predictive model.

Case Study 5 – Frac-hit Impact Quantification (Negative Impact)

A similar case is shown here as the previous case study (Figure 14), but this time around the frac-hit negatively impacts well performance. The vertical marker shows the time when the well suffered from frac-hit. The MS-DDV analysis allowed for estimation of average reservoir pressure and PI trends pre- and post-frac-hit. Each section is then individually forecasted using the same BHP profile that shows the expected profile of the well had it not seen frac-hit vs the actual behavior that shows significant underperformance

of 0.24 MMSTB cumulative oil volumes. This provides engineers to evaluate different strategies of frac-hit prevention or alleviate the negative impact by comparing the cases that showed an uplift in production because of frac-hit vs. the ones with negative impact.

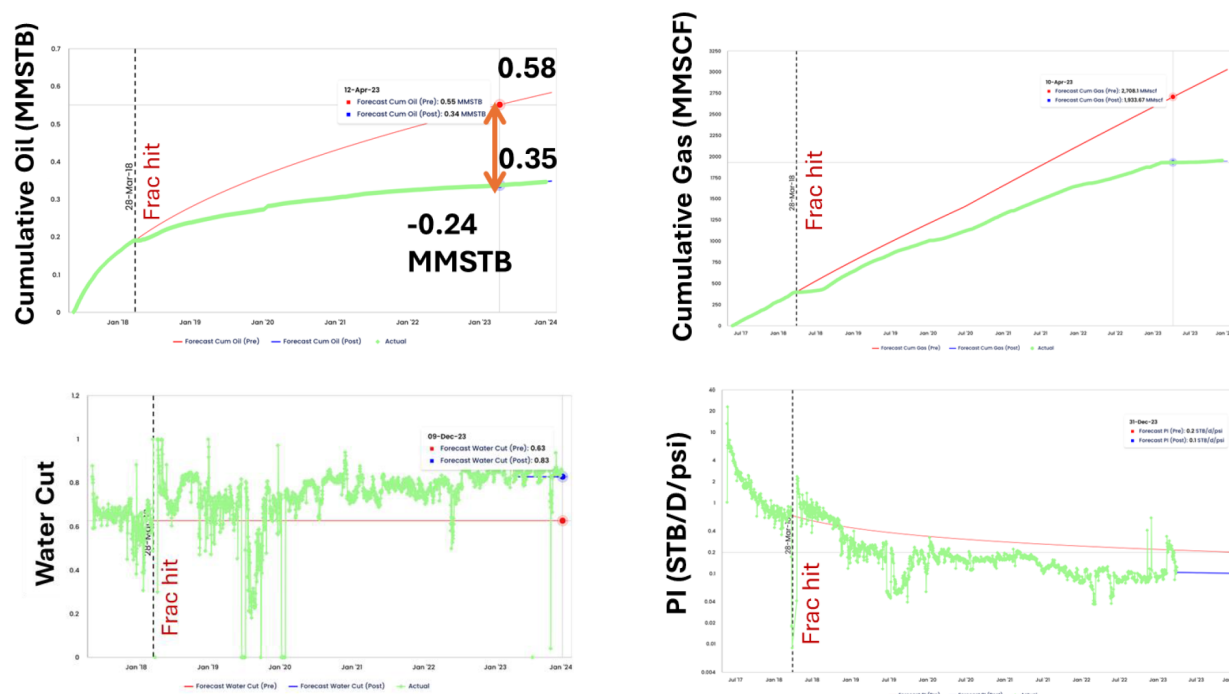


Figure 14—Single interference event (negative impact) – Multi segmented forecast of production rate, liquid productivity index, flowing bottom hole pressure and water cut

Table 5—Single interference event (negative impact) – Cumulative oil forward forecast for each segment (pre-frac hit and post-frac hit) using MS-PIBF

	Forecasted Cumulative Oil Volume (MMSTB)
Pre-Event	0.58
Post-Event	0.35
Difference	-0.24

Case Study 6 – Well Group/pad Study

Figure 15 illustrates the impact of frac-hit events because of a refrac in the neighboring well, using the pad B as an example. The well B5 was recently refrac'd, and the treatment had a measurable interference effect on nearby wells B4, B2, and B3. Following the refracture of B5, DDV for B4 and B2 showed a sudden decline, indicating drainage volume loss due to competitive fluid movement or pressure drawdown from the refrac well. B3 exhibited a slight DDV increase, suggesting limited positive stimulation from the event. The average reservoir pressure trends mirrored the DDV behavior, with B4 and B2 showing decreases and B3 showing a slight increase after the frac-hit. The flow regime classification from IRNP–MBT plots revealed distinct changes post-event, further confirming the impact of the frac-hit on reservoir connectivity and production dynamics. Such multi-well analysis is extremely valuable to compare and contrast well groups for future development decisions to maximize the production gains while minimizing such frac-hit impacts. The model is extremely fast to evaluate across the entire well fleet allowing for such indepth multi-well diagnostics.

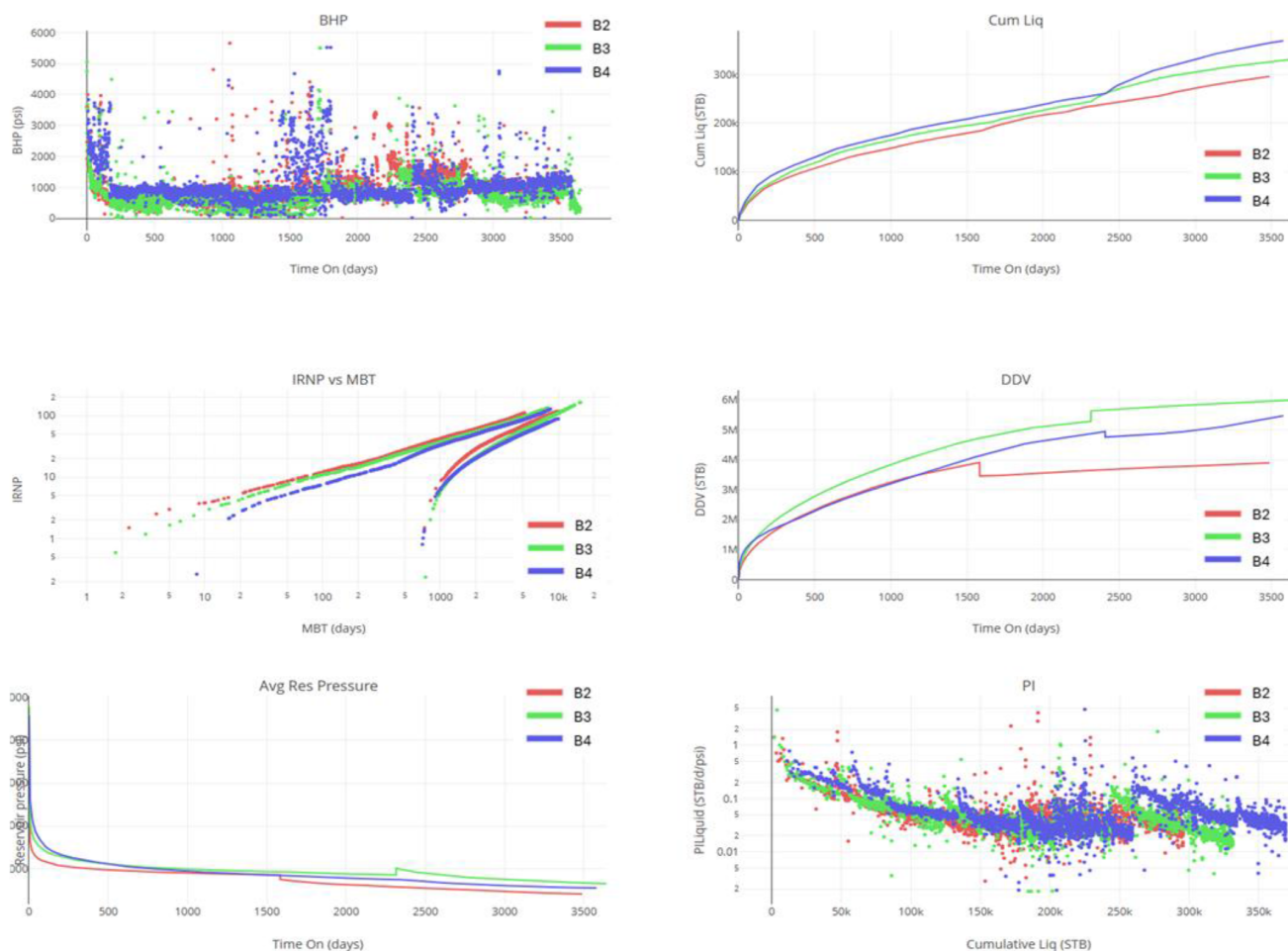


Figure 15—Well group/pad study

Conclusion

This study introduced a multi-segmented dynamic drainage volume (ms-DDV) that introduced two key ideas – 1. SINDy based data driven approach for automatic flow regime identification to aid accurate DDV estimates, and 2. A multi-segmented approach for DDV calculation to auto-identify the effect of extraneous factors such as frac-hit and refrac that fundamentally change the well behaviour and model the corresponding DDV profiles. With ms-DDV derived PI, PI based forecasting framework was used to detect, quantify, and forecast the impact of parent-child well interactions in unconventional reservoirs. By segmenting drainage volume evolution and applying sparse dynamical identification on each segment, the approach automatically captures performance discontinuities caused by events such as frac-hits, refracs, or stimulations and provides useful insights around the impact on each parameter due to the events. The methodology links changes in drainage volume directly to reservoir pressure, productivity index (PI), and rate forecasts, enabling a physics-consistent, data-efficient workflow deployable at field scale.

Field applications across multiple U.S. onshore basins demonstrated that the ms-DDV method can reliably identify both positive and negative interference effects, quantify their magnitude, and incorporate these shifts into accurate forward forecasts. Positive impacts were linked to fracture extension and repressurization, while negative impacts were tied to drainage area loss and accelerated depletion. The integration of ms-DDV with multi-segment PI-based forecasting (MS-PIBF) further improved predictive accuracy compared to single-segment methods. This framework offers operators a robust tool for real-time monitoring, performance diagnostics, and development planning, with direct applicability to refracture candidate screening and field-wide optimization strategies.

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